UNIVERSITY OF ASIA PACIFIC

Department of

Computer Science & Engineering

Project on

Bank Management System

Course No:CSE 201

Course Title:Object Oriented Programming II : Java

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Submitted To:	**Submitted By:**
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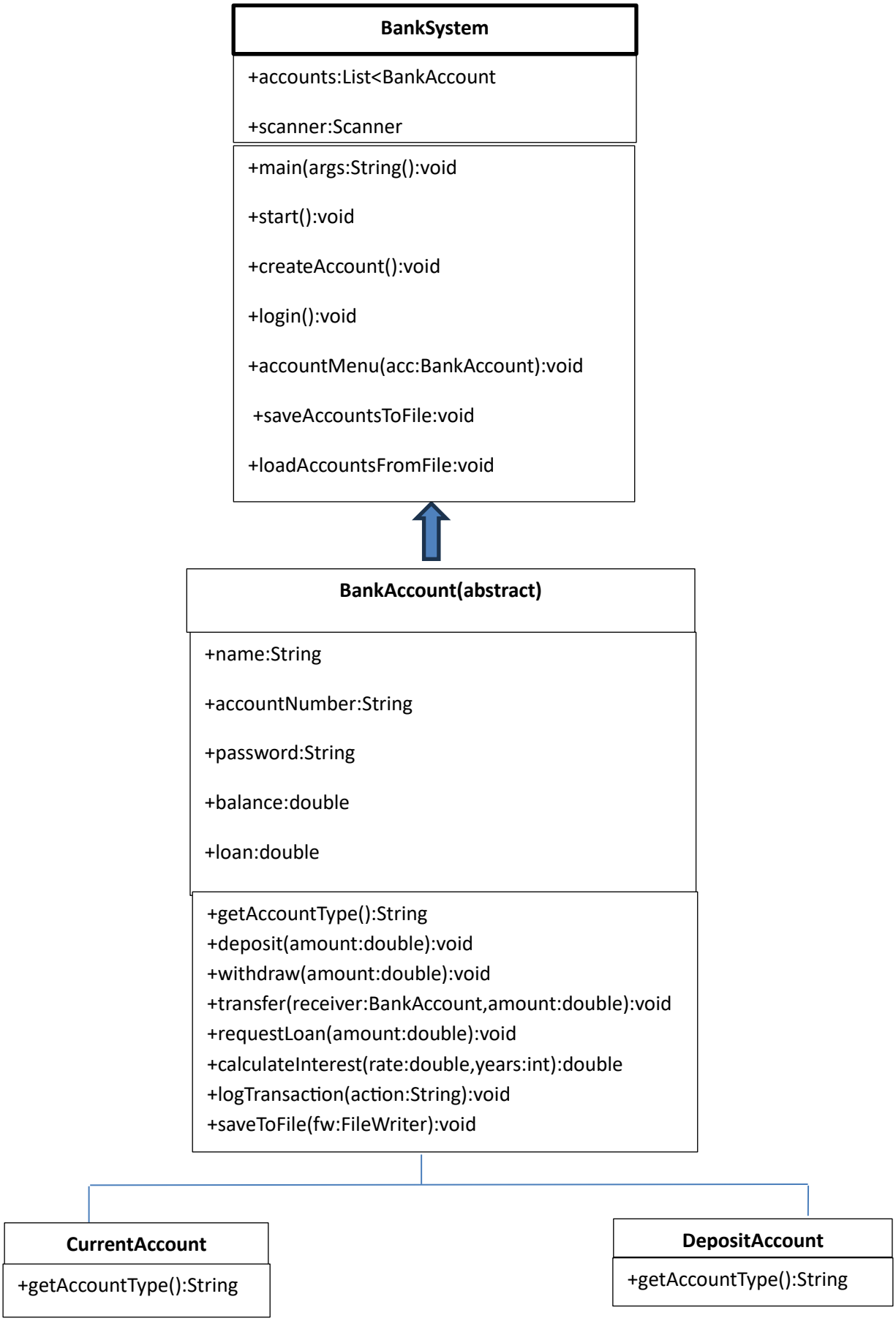
1. Short Introduction to the System:

This code is a simple Bank Management System in Java.Users can create accounts, log in, deposit, withdraw, and transfer money.There are two types of accounts: Current and Deposit.The system also supports loan requests and interest calculation (for deposit accounts).All account data is saved to a file and loaded when the program starts.

2. List of Identified Classes:

I. BankSystem: This is the main class that runs the app.It handles user input, account creation, login, and file saving/loading.
II. BankAccount (abstract):This is the base class for all accounts.It has common features like deposit, withdraw, transfer, and loan.
III. CurrentAccount:This class extends BankAccount for current account holders.It does not support interest calculation.
IV. DepositAccount:This class extends BankAccount for deposit account holders.It supports interest calculation using a fixed interest rate.

3.Class Diagram:



4.Code Implementation:

BankAccount(Abstract):

```
package project;


import java.io.FileWriter;

import java.io.IOException;


public abstract class BankAccount {


    private String name;

    private String accountNumber;

    private String password;

    protected double balance;

    private double loan;


    public BankAccount(String name, String accountNumber, String password, double balance) {

        this.name = name;

        this.accountNumber = accountNumber;

        this.password = password;

        this.balance = balance;

        this.loan = 0;

    }
```

```
public abstract String getAccountType();


public void deposit(double amount) throws IOException {
    balance += amount;
    logTransaction("Deposited TK" + amount);
}


public void withdraw(double amount) throws IOException {
    if (balance >= amount) {
        balance -= amount;
        logTransaction("Withdrawn TK" + amount);
    } else {
        throw new IOException("Insufficient balance.");
    }
}


public void transfer(BankAccount receiver, double amount) throws IOException {
    if (balance >= amount) {
        balance -= amount;
        receiver.balance += amount;
        logTransaction("Transferred TK" + amount + " to " + receiver.getAccountNumber());
        receiver.logTransaction("Received TK" + amount + " from " + this.accountNumber);
    } else {
```

```
            throw new IOException("Not enough balance to transfer.");
        }
    }

    public void requestLoan(double amount) throws IOException {
        loan += amount;
        balance += amount;
        logTransaction("Loan approved: TK" + amount);
    }

    public double calculateInterest(double rate, int years) {
        return (balance * rate * years) / 100;
    }

    protected void logTransaction(String action) throws IOException {
        FileWriter fw = new FileWriter("accounts.txt", true);
        fw.write("Account: " + accountNumber + ", Action: " + action + ", Balance: TK" + balance + "\n");
        fw.close();
    }

    // Updated to match the required format
    public void saveToFile(FileWriter fw) throws IOException {
        fw.write(name + "," + accountNumber + "," + password + "," + balance + "," + loan + ",0.0," +
getAccountType() + "\n");
```

```
    }

    public boolean checkPassword(String pass) {
        return password.equals(pass);
    }

    public String getName() {
        return name;
    }

    public String getAccountNumber() {
        return accountNumber;
    }

    public double getBalance() {
        return balance;
    }
}
```

CurrentAccount(Child Class):

```
package project;

public class CurrentAccount extends BankAccount {


    public CurrentAccount(String name, String accountNumber, String password, double balance) {

        super(name, accountNumber, password, balance);

    }


    @Override

    public String getAccountType() {

        return "Current";

    }

}
```

DepositAccount(Child Class):

```
package project;

public class DepositAccount extends BankAccount {

public DepositAccount(String name, String accountNumber, String password, double balance) {

        super(name, accountNumber, password, balance);

    }


    @Override

    public String getAccountType() {
```

```
        return "Deposit";
    }
}
```

BankSystem(Main Class):

```
package project;

import java.io.*;
import java.util.*;

public class BankSystem {
    private List<BankAccount> accounts = new ArrayList<>();
    private Scanner scanner = new Scanner(System.in);
    private final double INTEREST_RATE = 5.0;

    public static void main(String[] args) {
        BankSystem app = new BankSystem();
        app.loadAccountsFromFile(); // Load from text.txt
        app.start();
    }

    public void start() {
        while (true) {
```

```
        System.out.println("\n=== Bank System ===");
        System.out.println("1. Create Account");
        System.out.println("2. Login");
        System.out.println("3. Exit");
        System.out.print("Enter option: ");
        int choice = scanner.nextInt();
        scanner.nextLine();

        if (choice == 1) {
            createAccount();
        } else if (choice == 2) {
            login();
        } else if (choice == 3) {
            saveAccountsToFile(); // Save to text.txt
            System.out.println("Goodbye!");
            break;
        } else {
            System.out.println("Invalid option!");
        }
    }
}

private void createAccount() {
```

```
System.out.print("Name: ");
String name = scanner.nextLine();
System.out.print("Account Number: ");
String accNo = scanner.nextLine();

for (BankAccount b : accounts) {
    if (b.getAccountNumber().equals(accNo)) {
        System.out.println("Account with this number already exists!");
        return;
    }
}

System.out.print("Password: ");
String pass = scanner.nextLine();
System.out.print("Initial Deposit: ");
double deposit = scanner.nextDouble();
scanner.nextLine();
System.out.print("Account Type (Current/Deposit): ");
String type = scanner.nextLine();

BankAccount acc = null;
if (type.equalsIgnoreCase("Current")) {
    acc = new CurrentAccount(name, accNo, pass, deposit);
```

```
    } else if (type.equalsIgnoreCase("Deposit")) {
        acc = new DepositAccount(name, accNo, pass, deposit);
    }

    if (acc != null) {
        accounts.add(acc);
        saveAccountsToFile();
        System.out.println("Account Created and Saved!");
    } else {
        System.out.println("Invalid account type.");
    }
}

private void login() {
    System.out.print("Account Number: ");
    String accNo = scanner.nextLine();
    System.out.print("Password: ");
    String pass = scanner.nextLine();

    for (BankAccount acc : accounts) {
        if (acc.getAccountNumber().equals(accNo) && acc.checkPassword(pass)) {
            accountMenu(acc);
            return;
```

```
        }
    }
    System.out.println("Login failed.");
}

private void accountMenu(BankAccount acc) {
    while (true) {
        System.out.println("\nWelcome, " + acc.getName() + " [" + acc.getAccountType() + "]");
        System.out.println("1. Deposit");
        System.out.println("2. Withdraw");
        System.out.println("3. Transfer");
        System.out.println("4. Request Loan");
        System.out.println("5. Check Balance");
        System.out.println("6. Interest Calculation");
        System.out.println("7. Close Account");
        System.out.println("8. Logout");
        System.out.print("Option: ");
        int opt = scanner.nextInt();
        scanner.nextLine();

        try {
            if (opt == 1) {
                System.out.print("Amount (TK): ");
```

```
    acc.deposit(scanner.nextDouble());
    scanner.nextLine();
} else if (opt == 2) {
    System.out.print("Amount (TK): ");
    acc.withdraw(scanner.nextDouble());
    scanner.nextLine();
} else if (opt == 3) {
    System.out.println("Available Accounts:");
    for (BankAccount receiver : accounts) {
        if (!receiver.getAccountNumber().equals(acc.getAccountNumber())) {
            System.out.println("- " + receiver.getAccountNumber() + " (" + receiver.getName() + ")");
        }
    }
    System.out.print("Receiver Account Number: ");
    String receiverNo = scanner.nextLine();
    boolean found = false;
    for (BankAccount receiver : accounts) {
        if (receiver.getAccountNumber().equals(receiverNo)) {
            System.out.print("Amount to transfer (TK): ");
            acc.transfer(receiver, scanner.nextDouble());
            scanner.nextLine();
            found = true;
            break;
```

```
            }
        }
        if (!found) {
            System.out.println("Receiver not found.");
        }
    } else if (opt == 4) {
        System.out.print("Loan amount (TK): ");
        acc.requestLoan(scanner.nextDouble());
        scanner.nextLine();
    } else if (opt == 5) {
        System.out.println("Balance: TK" + acc.getBalance());
    } else if (opt == 6) {
        if (acc instanceof DepositAccount) {
            System.out.println("Interest Rate: " + INTEREST_RATE + "%");
            System.out.print("Years: ");
            int years = scanner.nextInt();
            scanner.nextLine();
            double interest = acc.calculateInterest(INTEREST_RATE, years);
            System.out.println("Interest: TK" + interest);
            System.out.println("Total: TK" + (acc.getBalance() + interest));
        } else {
            System.out.println("Only available for Deposit Accounts.");
        }
```

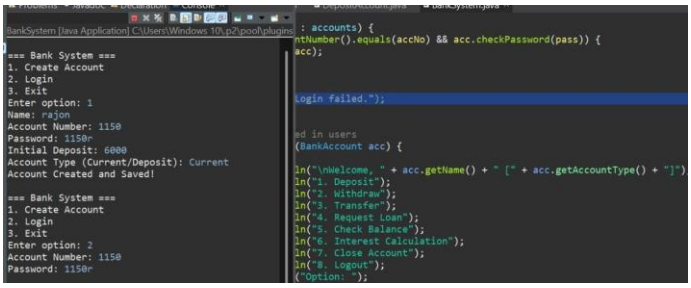
```
            } else if (opt == 7) {
                accounts.remove(acc);
                saveAccountsToFile();
                System.out.println("Account closed.");
                return;
            } else if (opt == 8) {
                saveAccountsToFile();
                return;
            } else {
                System.out.println("Invalid option!");
            }
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}

private void saveAccountsToFile() {
    try (FileWriter fw = new FileWriter("text.txt")) {
        for (BankAccount acc : accounts) {
            acc.saveToFile(fw);
        }
    } catch (IOException e) {
```

```
        System.out.println("Failed to save accounts.");
    }
}

private void loadAccountsFromFile() {
    try {
        File file = new File("text.txt");
        if (!file.exists()) return;

        Scanner fscan = new Scanner(file);
        while (fscan.hasNextLine()) {
            String[] parts = fscan.nextLine().split(",");
            if (parts.length >= 7) {
                String name = parts[0];
                String accNo = parts[1];
                String pass = parts[2];
                double balance = Double.parseDouble(parts[3]);
                String type = parts[6];

                BankAccount acc = null;
                if (type.equalsIgnoreCase("Current")) {
                    acc = new CurrentAccount(name, accNo, pass, balance);
                } else if (type.equalsIgnoreCase("Deposit")) {
```

```
                acc = new DepositAccount(name, accNo, pass, balance);
            }

            if (acc != null) {
                accounts.add(acc);
            }
        }
    }
    fscan.close();
  } catch (Exception e) {
    System.out.println("Error loading accounts.");
  }
 }
}
```

5.Screenshoot of output:

a. <u>Create and Login of current account:</u>



b. <u>Deposit and Check Balance of current account:</u>

```
Password: 1150r

Welcome, rajon [Current]
1. Deposit
2. Withdraw
3. Transfer
4. Request Loan
5. Check Balance
6. Interest Calculation
7. Close Account
8. Logout
Option: 1
Amount (TK): 500

Welcome, rajon [Current]
1. Deposit
2. Withdraw
3. Transfer
4. Request Loan
5. Check Balance
6. Interest Calculation
7. Close Account
8. Logout
Option: 5
Balance: TK6500.0

Welcome, rajon [Current]
```

c. <u>Logout of current account:</u>

```
Welcome, rajon [Current]
1. Deposit
2. Withdraw
3. Transfer
4. Request Loan
5. Check Balance
6. Interest Calculation
7. Close Account
8. Logout
Option: 8

=== Bank System ===
1. Create Account
2. Login
3. Exit
Enter option:
```

d. Create and Login of deposit account:

```
=== Bank System ===
1. Create Account
2. Login
3. Exit
Enter option: 1
Name: faria
Account Number: 1145
Password: 1145f
Initial Deposit: 9000
Account Type (Current/Deposit): Deposit
Account Created and Saved!

=== Bank System ===
1. Create Account
2. Login
3. Exit
Enter option: 2
Account Number: 1145
Password: 1145f

Welcome, faria [Deposit]
```

e. Withdraw and check balance of deposit account:

```
BankSystem [Java Application] C:\Users\Windows 10\.p2\pool\plugins
Password: 1145f

Welcome, faria [Deposit]
1. Deposit
2. Withdraw
3. Transfer
4. Request Loan
5. Check Balance
6. Interest Calculation
7. Close Account
8. Logout
Option: 2
Amount (TK): 5000

Welcome, faria [Deposit]
1. Deposit
2. Withdraw
3. Transfer
4. Request Loan
5. Check Balance
6. Interest Calculation
7. Close Account
8. Logout
Option: 5
Balance: TK4000.0

Welcome, faria [Deposit]
```

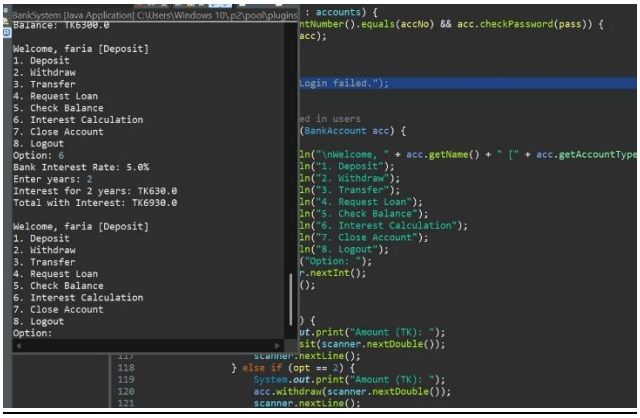
```
: accounts) {
ntNumber().equals(accNo) && acc.c
acc);

Login failed.");

ad in users
(BankAccount acc) {

ln("\nWelcome, " + acc.getName()
ln("1. Deposit");
ln("2. Withdraw");
ln("3. Transfer");
ln("4. Request Loan");
ln("5. Check Balance");
ln("6. Interest Calculation");
ln("7. Close Account");
ln("8. Logout");
("Option: ");
r.nextInt();
();

) {
ut.print("Amount (TK): ");
sit(scanner.nextDouble());
scanner.nextLine();
} else if (opt == 2) {
System.out.print("Amount (TK): ");
acc.withdraw(scanner.nextDouble());
```

f. Request for loan and check balance of deposit account:

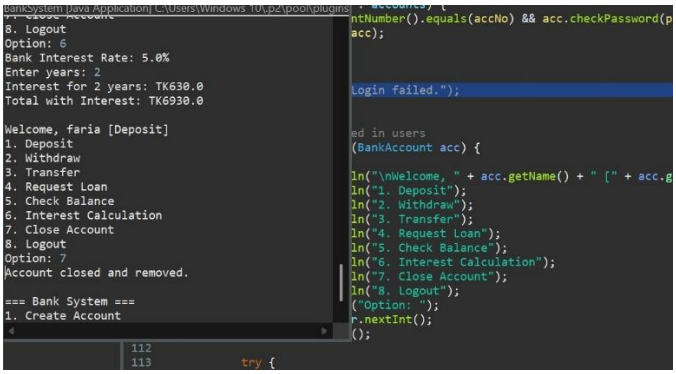
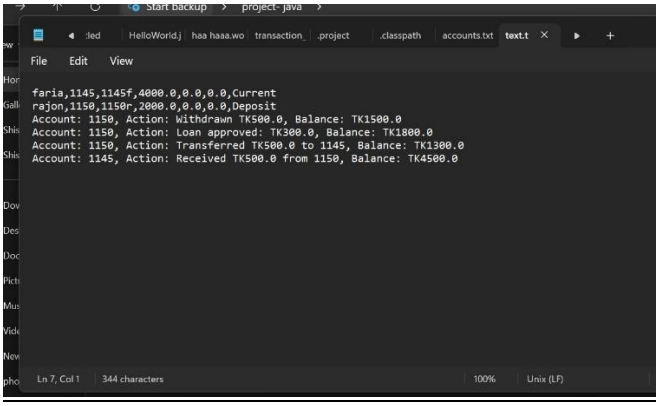
```
BankSystem [Java Application] C:\Users\Windows 10\.p2\pool\plugins
Balance: TK4000.0

Welcome, faria [Deposit]
1. Deposit
2. Withdraw
3. Transfer
4. Request Loan
5. Check Balance
6. Interest Calculation
7. Close Account
8. Logout
Option: 4
Loan amount (TK): 2300

Welcome, faria [Deposit]
1. Deposit
2. Withdraw
3. Transfer
4. Request Loan
5. Check Balance
6. Interest Calculation
7. Close Account
8. Logout
Option: 5
Balance: TK6300.0

Welcome, faria [Deposit]
```

```
: accounts) {
ntNumber().equals(accNo) && acc.checkPasswor
acc);

Login failed.");

ad in users
(BankAccount acc) {

ln("\nWelcome, " + acc.getName() + " [" + ac
ln("1. Deposit");
ln("2. Withdraw");
ln("3. Transfer");
ln("4. Request Loan");
ln("5. Check Balance");
ln("6. Interest Calculation");
ln("7. Close Account");
ln("8. Logout");
("Option: ");
r.nextInt();
();

) {
ut.print("Amount (TK): ");
sit(scanner.nextDouble());
scanner.nextLine();
```

g. Interest in deposit account:



h. Close account:



i.accounts info save in file: